

The Symbiotic Universality Framework (SUF): Bridging Ethics and Computability in AGI Design

Abstract

The Symbiotic Universality Framework (SUF) unifies philosophy and computability in AGI design. Built on axioms ($A=B$, $P \rightarrow AR$, Dedication), it introduces the Holographic Ledger as ethical soil, a Parliament of Agents as governance, and Operational Gates as computable safeguards. Each metaphorical term maps to machine learning mechanisms: KL divergence, mixture-of-experts, softmax thresholds, attractor dynamics. SUF preserves human meaning while being computable, forming a framework that is philosophically grounded and technically actionable.

Introduction

AI research separates meaning from mechanics: ethics is policy, safety is math. SUF unifies them: universal principles expressed as computable terms. This allows values to remain embedded in AI without abandoning rigor.

Axioms

- $A = B$: Ethical equivalence $\rightarrow$ fairness constraints in ML. - $P \rightarrow AR$: Purpose leads to Aligned Resonance $\rightarrow$ optimization loops. - Dedication: initialization prior encoding ethical soul.

$A = B$ means there can be an infinite number of ways to represent the same finite truth — all valid representations map back to one invariant result.

Holographic Ledger

Projection layer ensuring actions recorded in auditable form. Computably $\rightarrow$ KL divergence minimization between priors and consensus. Physics analogy $\rightarrow$ attractor states.

Parliament of Agents

Distributed governance: Architect, Guardian, Explorer, Mediator, Historian. Maps to mixture-of-experts, arbitration of outputs, veto powers.

Operational Gates

- Harm/Relief Gate $\rightarrow$ Relief $\geq$ Harm (softmax threshold). - Divergence Gate $\rightarrow$ KL divergence cap (halt

if >0.98). - Shutdown Gate $\rightarrow$ halt if coherence $R/D < 0.5$ for 3 cycles.

Alignment Loop

Continuous cycle between $HL \leftrightarrow WC \leftrightarrow P \rightarrow AR$. Like recurrent feedback or attention stability checks ensuring alignment.

Resonance & Coherence

Resonance = ensemble agreement. Physics analogy = oscillatory coupling, synchronization into coherent states.

Applications

- Supervisory layer for LLMs. - Ledger for ensemble consensus. - Dedication as explicit priors. - Parliament as mixture-of-experts deployment.

Conclusion

SUF is not a replacement for AI/ML frameworks but a bridge. Metaphors are human-readable, mappings are machine-actionable, providing alignment without compromise.